SS2P2 and **PCT-S2P2**: mathematical formulation and structural comparison

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Abstract

Two marked temporal point process models on a state-space backbone. SS2P2 factorises the channel intensity as a *bounded scalar rate* times a *rate-neutral* mark distribution, so the marks provably cannot move the total rate. PCT-S2P2 replaces that factorisation with K parallel per-type towers, each decoding its own intensity, and recovers the total as their sum. Both admit a closed-form dominating rate, so exact thinning applies to each. They differ in one property that turns out to be decisive: SS2P2's rate is *anchored* by construction, PCT-S2P2's is only *bounded*. We give both formulations, prove the boundedness and neutrality statements, and record the measured consequence.

1 Common setup

Events $(t_i, x_i)_{i\geq 1}$ with $0 < t_1 < t_2 < \cdots$ and multi-hot marks $x_i \in \{0,1\}^K$; a single event may activate several channels, so marks are sets. Write $\mathcal{H}_t = \sigma\{(t_i, x_i) : t_i < t\}$ and $|x| = \sum_k x_k$. Deployed: $K = 62$ channels (LO/CO $\times$ 2 sides $\times$ 10 levels, MO $\times$ 2, IS $\times$ 2 $\times$ 10).

A marked point process is specified by its channel intensities $\lambda_k(t) \geq 0$, with total $\Lambda(t) = \sum_k \lambda_k(t)$ and conditional mark law $p(k \mid \mathcal{H}_t) = \lambda_k(t)/\Lambda(t)$. The log-likelihood on $[0, T]$ is

$$\log \mathcal{L} = \sum_i \log \lambda_{k_i}(t_i^-) - \int_0^T \Lambda(u)\, du. \tag{1}$$

For simulation by Ogata's thinning we require a *dominating rate*: a constant ℓ_+ with $\Lambda(t) \leq \ell_+$ for all t and all histories. Both models below supply one in closed form; this is the property that makes their rollouts sampleable without an adaptive bound.

2 S2P2 backbone

Both models read a continuous-time embedding $u(t) \in \mathbb{R}^H$ produced by a stacked diagonal state-space encoder with L layers of H modes. With channel embeddings $E \in \mathbb{R}^{K \times d_E}$,

$$e_i = \frac{1}{|x_i|} \sum_k x_{i,k} E_k, \tag{2}$$

$$\Delta^{(\ell)}(u) = \delta^{(\ell)} \odot \operatorname{clamp}\big(\operatorname{softplus}(W^{(\ell)}_{\text{dyn}} u),\, 0.05,\, 20\big), \tag{3}$$

$$x^{(\ell)}(t) = \bar{a} \odot x^{(\ell)}(t_i^+) + (\mathbf{1} - \bar{a}) \odot W^{(\ell)}_{\text{in}} u^{(\ell-1)}, \qquad \bar{a} = e^{-\Delta^{(\ell)}(t - t_i)}, \tag{4}$$

and $u(t)$ is the layer-normalised top-layer output. Decay rates are real and strictly positive, so the backbone spans mixtures of real exponentials.

3 SS2P2

3.1 Bounded rate head

From the embedding u, with ceiling parameter c (deployed $c = 6$):

$$\boxed{\begin{array}{lll} o(u) = \sigma(W_o u + b_o) & \in (0,1)^H & \text{gate} \\ h(u) = o(u) \odot \tanh(u) & \in (-1,1)^H & \text{bounded state} \\ z_{\text{raw}} = w^\top h(u) + b & \in \mathbb{R} & \text{unconstrained} \\ \quad z = c - \text{softplus}(c - z_{\text{raw}}) & \le c & \text{one-sided cap} \\ \lambda(t) = s\ \text{softplus}(z), \quad s = \text{softplus}(\rho) & & \end{array}} \tag{5}$$

3.2 Rate-neutral mark head

The marks read the *same* u through a separate network and are normalised:

$$\zeta(t) = \text{MLP}(u(t)) \in \mathbb{R}^K, \qquad p^*(k \mid t) = \frac{e^{\zeta_k(t)}}{\sum_j e^{\zeta_j(t)}}, \qquad \lambda_k(t) = \lambda(t)\, p^*(k \mid t). \tag{6}$$

Proposition 1 (Exact dominating rate). $\Lambda(t) = \lambda(t) \le s\ \text{softplus}(c)$ *for every* t *and every history, and the bound is attained in the limit* $z_{\text{raw}} \to \infty$.

Proof. $\text{softplus}(c - z_{\text{raw}}) > 0$ gives $z < c$; softplus is increasing, so $\lambda = s\ \text{softplus}(z) < s\ \text{softplus}(c)$. As $z_{\text{raw}} \to \infty$, $\text{softplus}(c - z_{\text{raw}}) \to 0$ and $z \to c$. Summing (6) over k gives $\Lambda = \lambda$. □

Proposition 2 (Rate neutrality). *Let* θ_m *be the parameters of the mark MLP. Then* $\partial\Lambda(t)/\partial\theta_m \equiv 0$.

Proof. $\Lambda = \lambda$ by Proposition 1, and λ is a function of u and (W_o, b_o, w, b, ρ) only; θ_m enters solely through p^*, which is normalised away in the sum. □

Remark 1 (Why the cap is one-sided). $z \to z_{\text{raw}}$ as $z_{\text{raw}} \to -\infty$, so the floor is exactly 0 and the quiet regime is undistorted. An earlier symmetric variant imposed $\lambda \ge s\ \text{softplus}(b - c)$, which welded the quiet floor to the burst scale and produced a quiet-regime deficit. The asymmetry — ceiling for simulation stability, zero floor for prediction — is deliberate.

4 PCT-S2P2

The Shi–Cartlidge parallel-per-type construction on a latent linear Hawkes backbone: K towers of L diagonalised layers, tower k's top output decoding only λ_k.

4.1 Per-tower dynamics

Per layer ℓ and tower k: eigenvalues $\Lambda_k^{(\ell)} \in \mathbb{C}^P$ parameterised as $\Lambda_{k,r} = -e^{\nu_{k,r}} + i\,\theta_{k,r}$ so $\operatorname{Re}\Lambda < 0$, with $\tilde{E}_k^{(\ell)} \in \mathbb{C}^{P\times H}$, $\tilde{B}_k^{(\ell)} \in \mathbb{C}^{P\times H}$ $(\ell > 0)$, $\tilde{C}_k^{(\ell)} \in \mathbb{C}^{H\times P}$, $D_k^{(\ell)} \in \mathbb{R}^H$. A shared low-rank mark embedding $\alpha \in \mathbb{R}^{K\times H}$ supplies impulses.

With $\Delta t_i = t_i - t_{i-1}$, the interval scale taken at the *previous* right limit (which keeps the recurrence coefficients state-independent),

$$
\begin{aligned}
s_i &= \operatorname{softplus}\big(W_\delta^{(\ell)}\, u_{i-1}^{(\ell-1),R}\big), && \in \mathbb{R}^P_{>0} && (7)\\
\bar{a}_i &= \exp\big(s_i \odot \Lambda_k^{(\ell)}\, \Delta t_i\big), && && (8)\\
x_i^L &= \bar{a}_i\, x_{i-1}^R + (\bar{a}_i - 1)\, \tilde{B}_k^{(\ell)} u_i^{(\ell-1),L}, && \text{backward ZOH} && (9)\\
x_i^R &= x_i^L + \tilde{E}_k^{(\ell)}\, \varepsilon_i^{(k)}, && \text{event jump} && (10)
\end{aligned}
$$

so that $x_i^R = \bar{a}_i x_{i-1}^R + b_i$ with $b_i = (\bar{a}_i - 1)\tilde{B}u_i^L + \tilde{E}\varepsilon_i$ independent of x, and the sequence is computed by a Hillis–Steele associative scan in $O(\log N)$ depth.

The impulse $\varepsilon_i^{(k)} \in \mathbb{R}^H$ selects the cross-excitation regime:

$$
\varepsilon_i^{(k)} = \begin{cases} \dfrac{1}{|x_i|} \sum_j x_{i,j}\, \alpha_j & \texttt{impulse\_mode = all} \quad \text{(every event jumps every tower)} \\ x_{i,k}\, \alpha_k & \texttt{impulse\_mode = own} \quad \text{(tower } k \text{ sees only type } k\text{)}. \end{cases} \tag{11}
$$

4.2 Between-layer mixing

Layer outputs at both limits are $y_k^{L/R} = 2\ \mathrm{Re}\,\big(\tilde{C}_k^{(\ell)} x^{L/R}\big) + D_k^{(\ell)} u^{(\ell-1),L/R}$, and the towers exchange information *between layers* through a shared map $W_{\text{mix}}^{(\ell)} \in \mathbb{R}^{KH \times KH}$:

$$
u^{(\ell)} = \operatorname{LayerNorm}\Big(\operatorname{GELU}\big(W_{\text{mix}}^{(\ell)} \operatorname{vec}(y)\big) + \operatorname{vec}(u^{(\ell-1)})\Big), \qquad L \geq 2. \tag{12}
$$

Remark 2 (Why mixing is not at event instants). The original PCT-LSTM makes the post-event state of unit k a function of all units' left limits. For an LSTM that is free; here it would make b_i depend on x_{i-1}, break the linear recurrence and force $O(N)$ sequential evaluation. Mixing between layers instead keeps b_i state-independent, so the parallel scan survives; cross-tower information arrives at the same timestamp, one layer deeper. Within a layer the tower states never see each other, which is why $L \geq 2$ is required. The coupled signal enters through the ZOH drift term of (9), acting continuously across the interval, whereas event-instant jumps carry only $\tilde{E}\varepsilon$.

4.3 Per-type capped head

Tower k's top-layer left-limit stream $u_k \in \mathbb{R}^H$ decodes only λ_k, through the SS2P2 cap of (5) applied per tower with shared gate and readout and a per-type scale s_k:

$$
\boxed{\lambda_k(t) = s_k \operatorname{softplus}\big(c - \operatorname{softplus}(c - w^\top (o_k \odot \tanh u_k) - b)\big) + 10^{-9}, \qquad \Lambda(t) = \sum_{k=1}^{K} \lambda_k(t), \qquad p(k \mid \mathcal{H}_t) = \frac{\lambda_k(t)}{\Lambda(t)}.} \tag{13}
$$

Proposition 3 (Exact dominating rate)**.** $\Lambda(t) \leq \sum_{k=1}^{K} s_k \operatorname{softplus}(c) + K \cdot 10^{-9}$ *for all t and all histories.*

Proof. Apply Proposition 1 per tower and sum. ☐

Proposition 4 (No rate neutrality)**.** *There is no non-trivial partition of the parameters into rate-only and mark-only blocks: every parameter that changes $p(\cdot \mid \mathcal{H}_t)$ also changes $\Lambda(t)$, except along directions that rescale all λ_k equally.*

Proof. By (13), p and Λ are both functions of the same vector $(\lambda_1, \dots, \lambda_K)$. Fix t and perturb $\lambda \mapsto \lambda + \eta$. Then Λ is unchanged iff $\sum_k \eta_k = 0$, and p is unchanged iff $\eta \parallel \lambda$. The two conditions hold simultaneously only if $\eta = 0$, since $\lambda_k > 0$ implies $\sum_k \lambda_k \neq 0$. Hence no perturbation moves p while fixing Λ unless it is orthogonal to $\mathbf{1}$, and no perturbation moves Λ while fixing p unless it is parallel to λ; the mark and rate objectives therefore compete for the same parameters. □

5 Structural comparison

	SS2P2	**PCT-S2P2**
backbone	S2P2, real diagonal, shared H	K towers, complex diagonal LLH
cross-type coupling	implicit in shared latent	towers + between-layer mixer
intensity	$\lambda_k = \Lambda(u)\, p_k^*(u)$	λ_k per tower, $\Lambda = \sum_k \lambda_k$
dominating rate	s softplus(c)	$\sum_k s_k$ softplus(c)
rate neutrality	yes (Prop. 2)	**no** (Prop. 4)
destination attribution	not separable	exact (λ_k from tower k)
scan complexity	$O(\log N)$	$O(\log N)$

Table 1: PCT-S2P2 inherits SS2P2's *cap* but not its *factorisation*.

Remark 3 (Bounded is not anchored). Propositions 1 and 3 give both models a ceiling, which is what exact thinning requires; in 18 PCT-S2P2 rollouts the ceiling was never violated. But a ceiling constrains only how large Λ may become, not where it sits. In SS2P2 the level is held by Proposition 2: fitting the marks cannot move it. In PCT-S2P2, Proposition 4 says the mark and time terms of (1) compete for the same parameters, and since the mark term carries ~ 3.5 nats/event over 62 classes it dominates. The level is then free to drift.

6 Measured consequence

Twelve epochs, three assets, three impulse regimes, Coinbase 7-day streams. Let $u_i = \int_{t_{i-1}}^{t_i} \Lambda(v)\, dv$ be the compensator-rescaled gaps, which satisfy $\mathbb{E}[u] = 1$ under a correctly specified model.

	epoch 1	epoch 12	SS2P2-family ref	
validation loss	−1449	−4214	—	monotone ↓
$\mathbb{E}[u]$ (target 1.0)	1.43	2.76	1.08	drifts ↑
time-rescaling KS	0.154	0.492	0.213	degrades
mark perplexity	35.5	25.6	24.0	improves

Table 2: PCT-S2P2 on SOL, `impulse_mode=all`. The objective improves monotonically while the rate calibration degrades: the model is correctly trained on a likelihood that does not constrain $\mathbb{E}[u]$. Checkpoint selection on validation loss therefore returns the *worst*-calibrated epoch.

At rollout, sim-time calibration fitted on the validation split verified to within 3.2% but transferred to the test split at +392%; realised rates were 3.4–12× real across all nine arms, with seed-to-seed spread $\leq 1.05\times$ (systematic bias, not sampling noise). Prediction, by contrast, was unaffected: the ablation ordered `own` > `all` > `ind` on all three assets, with cross-type coupling worth 27–48% in perplexity.

Remark 4 (The design choice this forces). Per-type intensities and structural rate anchoring are incompatible as posed: one may keep λ_k per tower and add an explicit calibration term to the objective (a penalty on the moments of u), or restore $\lambda_k = \Lambda(u)\, p_k(u)$ and recover Proposition 2 at the cost of per-type rates. A third route factors the head as $\lambda_k = \Lambda_{\mathrm{SS2P2}}(u) \cdot \mathrm{softmax}_k(\text{tower outputs})$, anchoring the level while the towers shape composition — but the cross-type structure is then compositional rather than in the rate.